

TANMAY SINGH

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EDUCATION

SRM Institute of Science and Technology

2024 – 2028

B. Tech in Computer Science and Engineering, Cloud Computing Specialization | CGPA: 9.53/10

WORK EXPERIENCE

Research Intern

Apr 2026 – Present

Samsung PRISM — Context-Aware Quick Control for IoT Devices

- Engineered a multi-sensor fusion pipeline combining YOLOv8n object detection, BLE proximity, and IMU data to drive context-aware IoT device selection on Android.
- Architected BLE scanning, CameraX vision, and Home Assistant REST integration modules with Room Database synchronization, using Kotlin, Jetpack Compose, and Coroutines.

Backend Engineering Intern

Dec 2025 – Jan 2026

MedGency

- Designed and implemented an authentication and authorization architecture for a healthcare platform, including RBAC, JWT, and OTP-based verification, using Node.js, Express.js, and MongoDB.
- Built invite-based onboarding, organization isolation, audit logging, and Redis-backed token revocation to enforce fail-closed, Zero Trust access control.

PROJECTS

CodeWeave

GitHub

AI-Assisted Real-Time Collaborative Coding Platform

React, Node.js, Express.js, Socket.IO, MongoDB, Redis

- Architected a real-time collaborative coding platform using Socket.IO, MongoDB, and Redis, enabling synchronized code editing, shared workspaces, JWT-based authentication, and low-latency client-server communication.
- Integrated Google Gemini AI, the Monaco Editor, REST APIs, and WebSocket's to deliver AI-assisted code generation, collaborative messaging, online presence tracking, and secure project-based collaboration.

Self-Healing Database Engine

GitHub

Database-Native Reliability & Self-Healing System

MySQL, Stored Procedures, FastAPI, Next.js

- Designed a database-native self-healing engine where anomaly detection, decision-making, remediation, and verification execute directly inside MySQL using Stored Procedures and the Event Scheduler.
- Implemented statistical anomaly detection using time-weighted baselines, Log-Normalized Z-Score, IQR, and Coefficient of Variation, with a FastAPI backend and Next.js dashboard for monitoring and

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Tailwind CSS

Databases: MongoDB, MySQL, Redis, SQLite

Cloud & DevOps: Docker, Kubernetes, Git, GitHub

Developer Tools: Postman, VS Code, IntelliJ IDEA

Computer Science: Data Structures & Algorithms, OOP, DBMS, Operating Systems

ACHIEVEMENTS

- Samsung PRISM Research Program** – Selected for research on **Context-Aware IoT Systems** using Computer Vision and Android.
- OSSome Hacks 2.0** – Awarded **Best Freshers Team** for developing **Samvidhan Setu**, an AI-powered legal research platform.
- Java Project Expo 2025, SRMIST** – Awarded **3rd Prize** for **FileX**, an enterprise endpoint telemetry and threat detection system.